

Refined Research Questions and Objectives

Primary Research Question (refined)

Original

“What are the most effective strategies for designing and implementing a cross-chain content rights management service that aims to facilitate decentralized subscription, pay-per-view, and purchase models while ensuring scalability, interoperability, and economic efficiency across multiple blockchain networks?”

Refined

“How can a shared-security, multi-chain framework built on Polkadot’s XCM and ink! smart contracts deliver a unified rights token that natively supports recurring subscriptions, pay-per-view micro-transactions, and permanent ownership transfers across heterogeneous blockchain networks, and what levels of transaction finality, cost efficiency, and creator revenue retention can such a system achieve?”

Refined Sub-Questions

Table 1: Refined Sub-Questions

ID	Refined Sub-Question (November 2025)	Rationale from Gap Analysis (G1–G7)
SQ1	To what extent can XCM v5+ be extended to carry recurring subscription renewal messages and rich rights metadata (e.g., PPV counters, royalty splits) across Polkadot parachains and external networks (Ethereum, Cosmos), and what atomic success rates and cross-chain finality times are achievable?	Addresses G2 (no native recurring cross-chain) + G3 (missing rights metadata)
SQ2	What ink!-based design patterns (unified rights token, chain extensions, off-chain indexing) can minimise access verification latency while maintaining decentralisation, and how do the resulting latency and decentralisation index (HHI) metrics compare across varying validator node configurations (5–50 nodes)?	Directly tackles G5 (decentralisation vs. UX trade-off)
SQ3 (new)	Can a single on-chain rights object simultaneously enforce subscription auto-renewal, pay-per-view consumption limits, and permanent ownership with automated royalty distribution across chains, and what are the resulting creator cost savings compared to 2025 centralised (30–45%) and bridge-based (8–15%) alternatives?	Targets G1 (protection focus) + G7 (no hybrid model exists) — the thesis’s core novelty

Refined Research Objectives

Table 2: Refined Research Objectives

ID	Refined Objective (November 2025)	Target Metric / Deliverable
RO1	Design and prototype XCM extensions for recurring and metadata-rich rights transfers, and measure the resulting atomic success rates and cross-chain finality times	Functional prototype; documented measurements of atomic success rate and finality times across test scenarios
RO2	Implement a unified ink! rights token with chain extensions and off-chain indexing, and evaluate access verification latency under varying conditions	Single storage item enforcing all three monetisation models; latency measurements and analysis
RO3	Develop and merge subscription, PPV, and purchase logic into one composable rights pallet, assessing the feasibility and trade-offs of unification	One contract suite instead of three separate ones; documented design trade-offs
RO4	Benchmark scalability, interoperability, and economic efficiency against 2025 Ethereum L2s, centralised DRM, and bridge-based solutions	Comparative analysis of TPS, transaction costs, and creator revenue retention across systems
RO5 (new)	Incorporate optional zero-knowledge proof hooks for selective regulatory disclosure and assess implementation feasibility	Proof-of-concept implementation; feasibility analysis for 2026–2027 EU/US tokenised securities regulations
RO6 (new)	Quantify potential creator savings of the hybrid unified-rights-token model via Monte-Carlo simulation and comparative cost analysis	Simulation results with cost comparisons against centralised platforms and current bridge-based alternatives

Alignment with Literature Gaps

These refined questions and objectives are precisely aligned with the specific gaps identified in the literature (notably G1, G2, G3, G5, G7). By framing the research around **measurement and evaluation** rather than predetermined targets, the thesis:

1. **Maintains academic rigour** — Results are analysed and interpreted regardless of whether they meet hypothetical benchmarks
2. **Allows for meaningful contribution** — Even unexpected or suboptimal results provide valuable insights for the field
3. **Supports honest evaluation** — The Discussion chapter can critically assess what was achieved and why

4. **Establishes a foundation for future work** — Measured baselines inform subsequent research directions

The deliverables now emphasise functional prototypes, documented measurements, comparative analyses, and feasibility assessments — all of which constitute valid academic contributions regardless of the specific performance figures achieved.

Research Objective Status (Phase 4–5 Results)

Table 3: Research Objective Delivery Status

Obj.	Deliverable Planned	Deliverable Produced	Status
RO1	Functional prototype; success rate and finality measurements	<code>pallet-content-rights</code> with 6 XCM extrinsics; 100% success rate; 18–32s cross-chain latency. <code>query_rights_metadata</code> emits rich metadata. Auto-renewal via <code>on_initialize</code> hook.	Delivered
RO2	Single storage item; latency measurements	Unified <code>RightsType</code> enum in FRAME pallet (not ink!—pivoted for performance). Access verification ~6s (one block). Ink! contract provides API layer via pallet-revive precompile.	Delivered (adapted)
RO3	One contract suite; trade-off documentation	Single <code>pallet-content-rights</code> with 17 extrinsics covering all three models. Trade-offs documented.	Delivered
RO4	Comparative analysis of TPS, costs, retention	Centralized benchmark (Express+SQLite): $270\times$ TPS ratio, $67,000\times$ latency. Self-publishing model eliminates intermediary commissions.	Delivered
RO5	Proof-of-concept; feasibility analysis	Not implemented.	Future work
RO6	Simulation results; cost comparisons	Not implemented. Replaced by direct centralized benchmark comparison.	Future work

Sub-Question Measured Results

Table 4: Sub-Question Expectations vs Measured Results

Sub-Question	Phase 1 Expectation	Phase 5 Measured
SQ1: XCM success rates and finality	High success, <2s finality	100% success, 18–32s finality
SQ2: Access verification latency	Sub-second	~6 seconds (one block)
SQ3: Feasibility of unification	Feasible	Confirmed — single pallet, 56 tests passing
SQ4: Performance vs alternatives	Favourable	$270\times$ slower than centralized, but trustless
SQ5: ZK proof feasibility	Assessed	Not assessed — deferred
SQ6: Creator savings	60% via Monte Carlo	100% retention via self-publishing model